PCBC H Name _____ _____ /22

Limits Quiz 1.1–1.2

Non-Calculator

You have 20 minutes.

PRACTICE A · built to match the real Quiz 1.1–1.2: 11 questions, 20 minutes, no calculator

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.

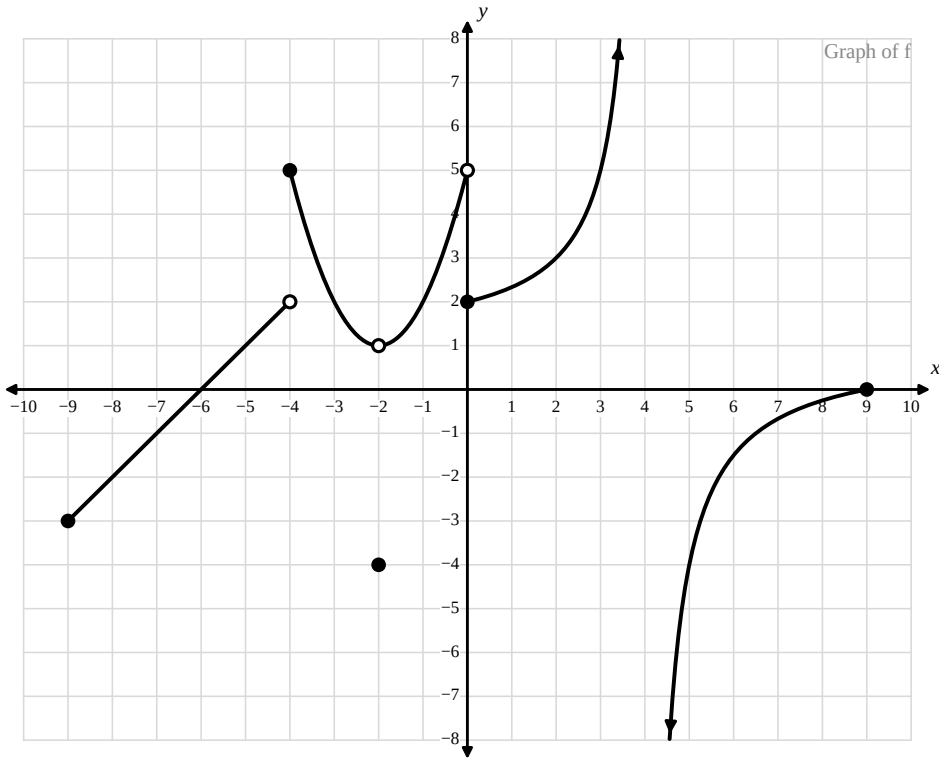
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Signature _____

SHOW ALL WORK FOR FULL CREDIT.

Use the graph of f shown below to find each. If a limit does not exist, write DNE. (2 points each)

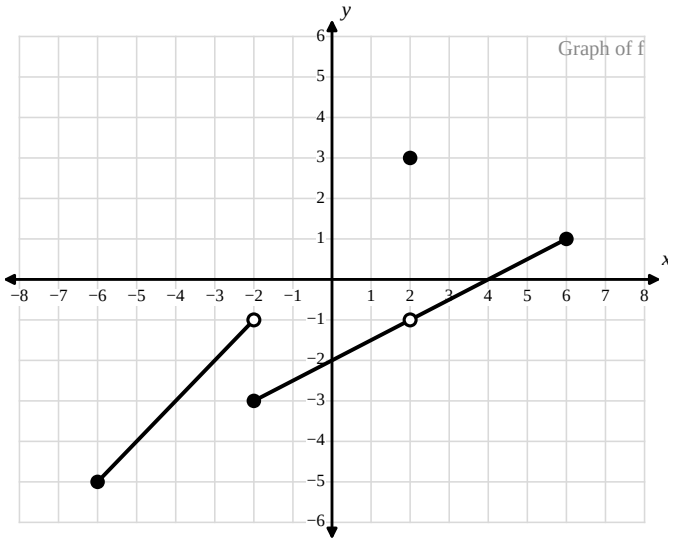


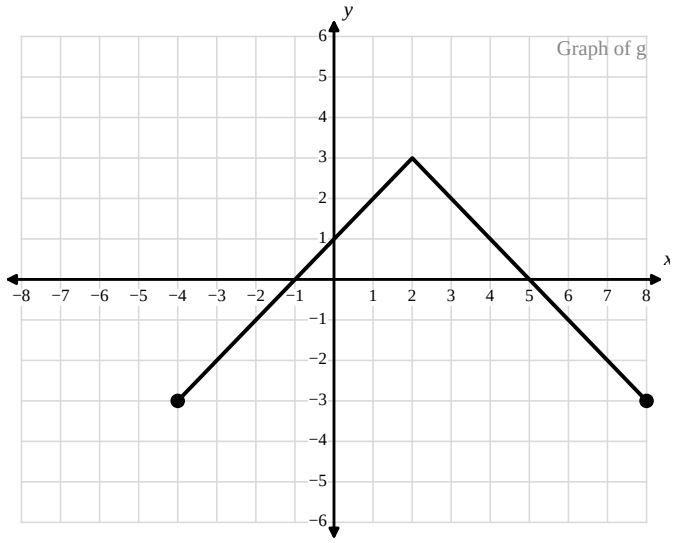
1. $\lim_{x\to -2} f(x) =$	**2.** $\lim_{x\to -4} f(x) =$	**3.** $\lim_{x\to 4^+} f(x) =$

(2 points)

4. Use the graph of f above to determine all values of a for which $\lim_{x\to a} f(x) = 2$ on the interval $-9 < x < 9$. If there are no values of a, write the empty set.

Use the graphs of f and g below to find the limit. Show the limit properties you use. (2 points)





5. $\lim_{x \to -2^+} [3f(x) - g(x)]$

(2 points)

6. Use the table to estimate $\lim_{x \to 3} g(x)$. Give the left-hand limit, the right-hand limit, and the limit, or state that it does not exist.

x	2.9	2.99	2.999	$\to 3 \leftarrow$	3.001	3.01	3.1
$g(x)$	0.2515823	0.2501564	0.2500156		0.2499844	0.2498439	0.2484567

$\lim_{x \to 3^-} g(x) \approx$ _____ $\lim_{x \to 3^+} g(x) \approx$ _____ $\lim_{x \to 3} g(x) \approx$ _____

Evaluate each limit. (2 points each)

7. $\lim\limits_{x \to 3^-} \dfrac{x-3}{|x-3|}$

8. $\lim\limits_{x \to -3} \dfrac{x^2+x-6}{x^2-9}$

9. $\lim\limits_{x \to 2} \dfrac{x-2}{\sqrt{2x+5}-3}$

10. $\lim\limits_{x \to 0} \dfrac{\frac{1}{x+3} - \frac{1}{3}}{x}$

11. For $f(x) = x^2 - 5x + 2$, find $\lim\limits_{h \to 0} \dfrac{f(x+h) - f(x)}{h}$

Each step: the work on the left, the reason on the right. "lim" stays on every line until the line where you substitute.

1. $\boxed{1}$

Both sides of the curve head to the open circle at $(-2, 1)$.	The filled dot at $(-2, -4)$ is $f(-2)$; a limit ignores the point at $x = -2$.

2. $\boxed{\text{DNE}}$

$\lim_{x \to -4^-} f(x) = 2$, $\lim_{x \to -4^+} f(x) = 5$	From the left the line ends at the open circle $(-4, 2)$; from the right the curve starts at $(-4, 5)$.
$2 \neq 5$, so $\lim_{x \to -4} f(x)$ does not exist.	Two-sided needs both sides to match.

3. $\boxed{-\infty}$

Just right of $x = 4$ the curve falls without bound.	One-sided limit from the right only. (From the left it rises to $+\infty$.)

4. $\boxed{a = -3,\ a = -1}$

The curve reaches height 2 from both sides at $x = -3$ and $x = -1$.	Two-sided: approach $y = 2$ from the left and the right.
Not $x = -4$: only the left side heads to 2 (the right side starts at 5).	A one-sided approach does not count.
Not $x = 0$: only the right side starts at 2 (the left side heads to 5).	Same reason.

5. $\boxed{-8}$

$\lim_{x \to -2^+} [3f(x) - g(x)] = 3 \lim_{x \to -2^+} f(x) - \lim_{x \to -2^+} g(x)$	Constant multiple and difference properties.
$= 3(-3) - (-1)$	From the right, f starts at the filled dot $(-2, -3)$; $g(-2) = -1$.
$= -8$	Trap: the left side of f heads to -1, which would give -2.

6. $\boxed{\approx 0.25}$

Left: 0.2515823, 0.2501564, 0.2500156	Closest value rounds to 0.250.
Right: 0.2484567, 0.2498439, 0.2499844	Closest value rounds to 0.250.
$\lim_{x \to 3} g(x) \approx 0.25$	Both sides agree. A table estimates; it does not prove.

7. $\boxed{-1}$

$\lim_{x \to 3^-} \dfrac{x - 3}{\lvert x - 3 \rvert} = \lim_{x \to 3^-} \dfrac{x - 3}{-(x - 3)}$	For $x < 3$, $x - 3$ is negative, so $\lvert x - 3 \rvert = -(x - 3)$.
$= \lim_{x \to 3^-} (-1) = -1$	Cancel; limit of a constant.

8. $\boxed{\frac{5}{6}}$

Substitution produces 0/0: indeterminate.	Factor.
$\lim\limits_{x\to -3}\dfrac{x^2+x-6}{x^2-9}=\lim\limits_{x\to -3}\dfrac{(x+3)(x-2)}{(x+3)(x-3)}$	Trinomial and difference of squares.
$=\lim\limits_{x\to -3}\dfrac{x-2}{x-3}$	Cancel $x+3$; allowed because $x\neq -3$.
$=\dfrac{-3-2}{-3-3}=\dfrac{-5}{-6}=\dfrac{5}{6}$	Substitute.

9. $\boxed{3}$

Substitution produces 0/0: indeterminate.	Radical: multiply by the conjugate.
$\lim\limits_{x\to 2}\dfrac{x-2}{\sqrt{2x+5}-3}=\lim\limits_{x\to 2}\dfrac{(x-2)\left(\sqrt{2x+5}+3\right)}{(2x+5)-9}$	Bottom is a^2-b^2; leave the top factored.
$=\lim\limits_{x\to 2}\dfrac{(x-2)\left(\sqrt{2x+5}+3\right)}{2(x-2)}$	$2x-4=2(x-2)$.
$=\lim\limits_{x\to 2}\dfrac{\sqrt{2x+5}+3}{2}$	Cancel $x-2$.
$=\dfrac{\sqrt{9}+3}{2}=\dfrac{6}{2}=3$	Substitute.

10. $\boxed{-\frac{1}{9}}$

Substitution produces 0/0: indeterminate.	Complex fraction: multiply top and bottom by the LCD $3(x+3)$.
$\lim\limits_{x\to 0}\dfrac{\frac{1}{x+3}-\frac{1}{3}}{x}=\lim\limits_{x\to 0}\dfrac{3-(x+3)}{3x(x+3)}$	$3(x+3)\cdot\frac{1}{x+3}=3$ and $3(x+3)\cdot\frac{1}{3}=x+3$.
$=\lim\limits_{x\to 0}\dfrac{-x}{3x(x+3)}$	Distribute the negative: $3-x-3=-x$.
$=\lim\limits_{x\to 0}\dfrac{-1}{3(x+3)}$	Cancel x.
$=\dfrac{-1}{3(0+3)}=-\dfrac{1}{9}$	Substitute.

11. $\boxed{2x-5}$

$\lim\limits_{h\to 0}\dfrac{\left[(x+h)^2-5(x+h)+2\right]-\left(x^2-5x+2\right)}{h}$	Build $f(x+h)$ with brackets.
$=\lim\limits_{h\to 0}\dfrac{x^2+2xh+h^2-5x-5h+2-x^2+5x-2}{h}$	Expand; distribute the negative.
$=\lim\limits_{h\to 0}\dfrac{h(2x+h-5)}{h}$	Combine; factor out h.
$=\lim\limits_{h\to 0}(2x+h-5)=2x+0-5=2x-5$	Cancel h; substitute 0 for h, not x.